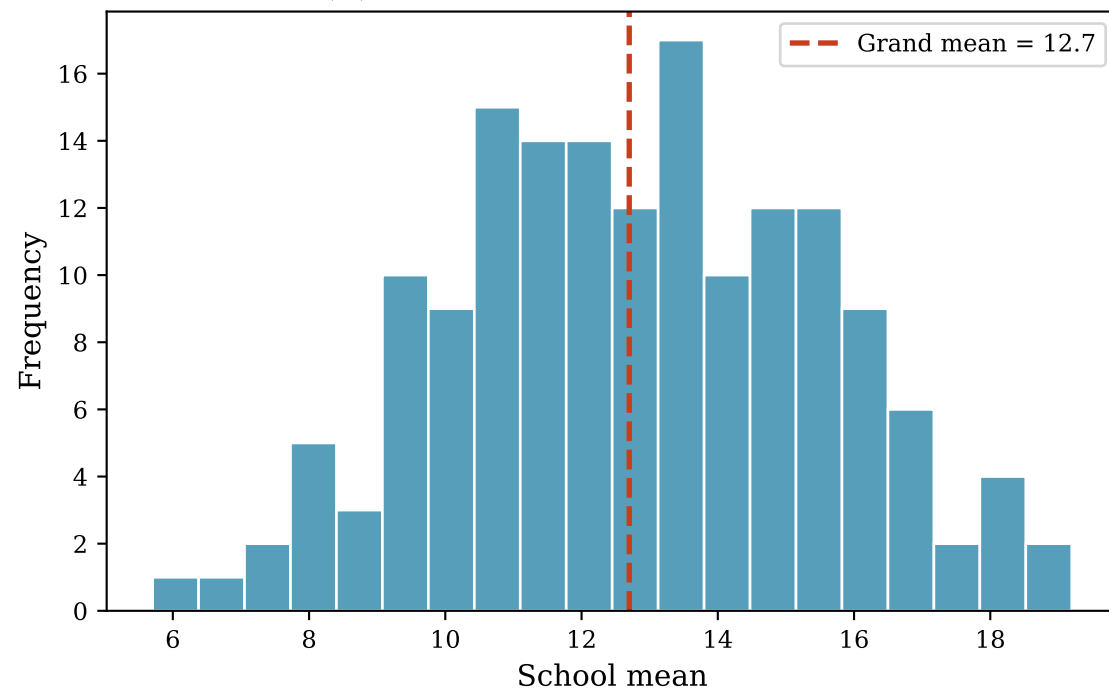
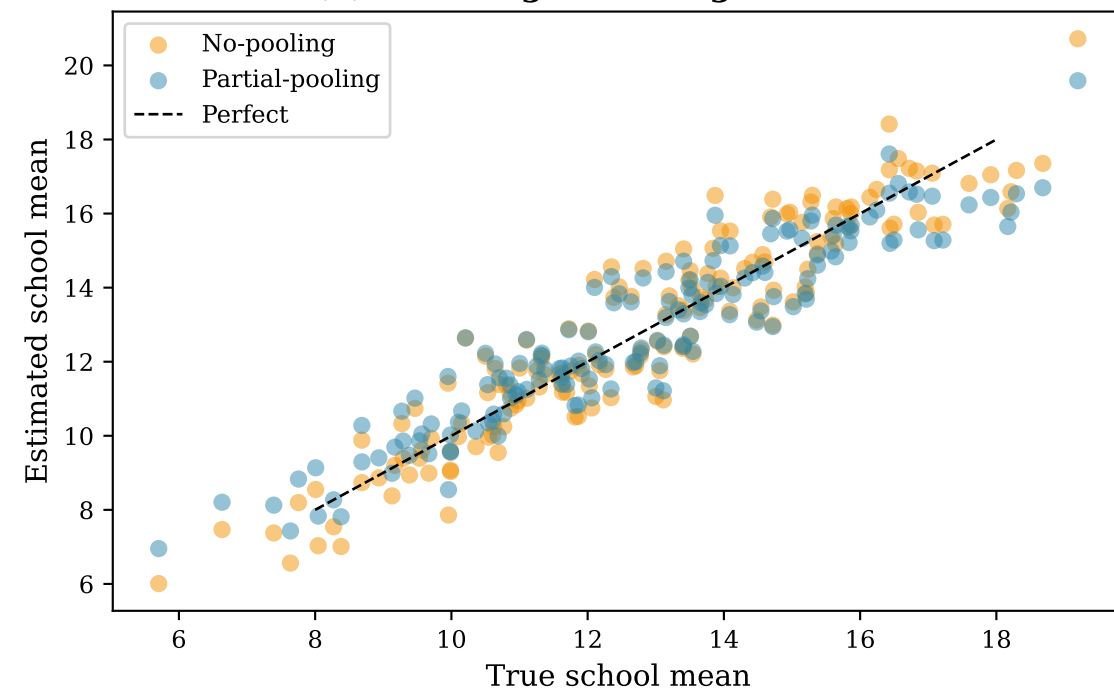


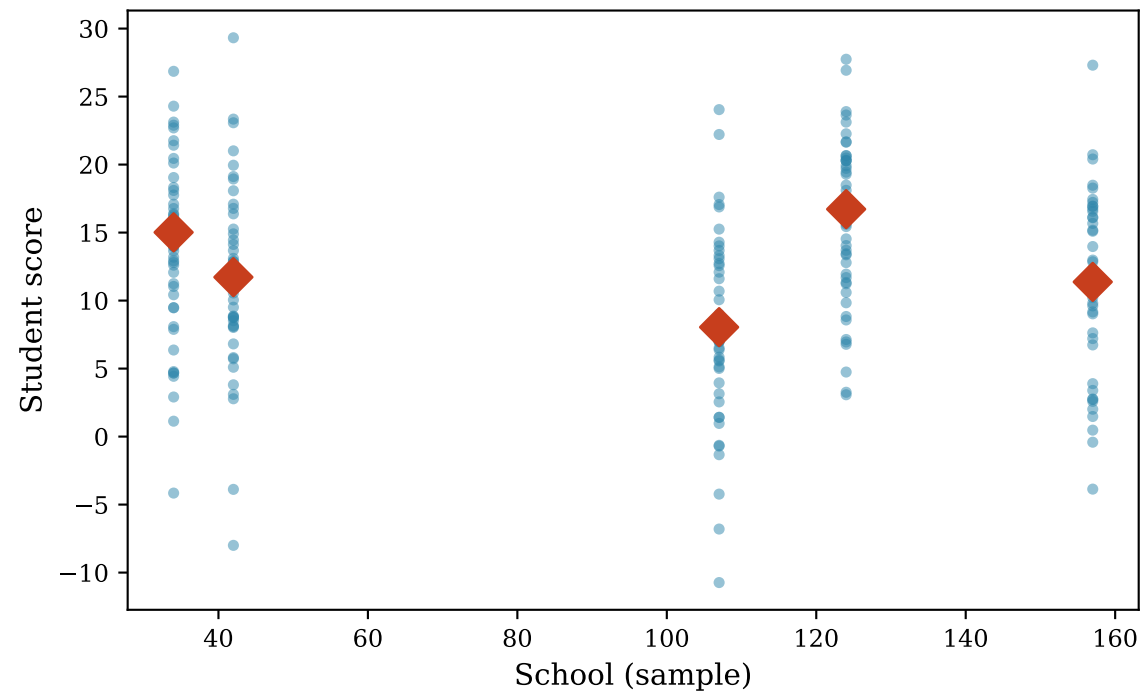
(A) Distribution of school means



(C) Shrinkage toward grand mean



(B) Within-school variation



HSB-Calibrated Simulation

Schools: 160
Students per school: 45

Grand mean: 12.7
Between-school SD (τ): 2.5
Within-school SD (σ): 6.8

ICC: 0.119
Reliability: 0.859
CF = Reliability: 0.859

Recommendation: No-pooling acceptable